US Green Energy Transition: Visualizing the Past, Present, and Future

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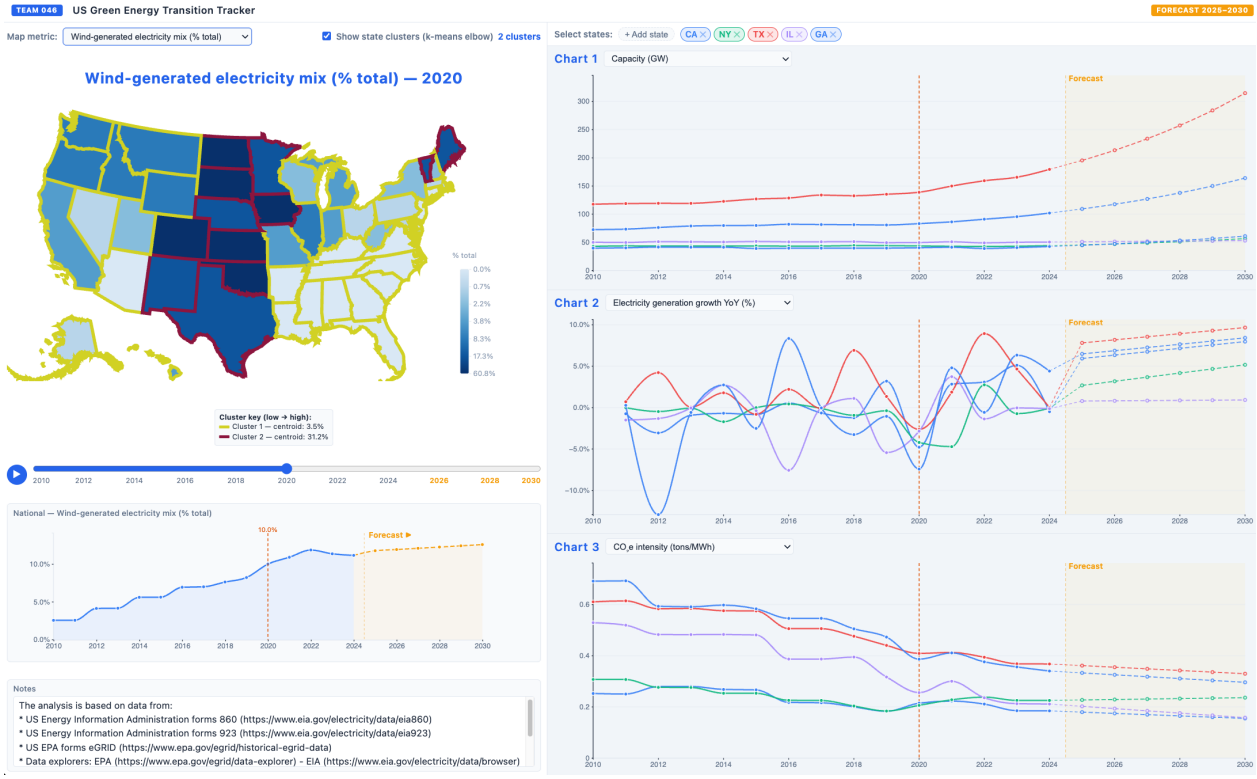
Summary

The US electricity grid is experiencing a large shift in demand. Higher needs from AI datacenters, retirements of coal power plants, and investment in renewables are all changing electricity capacity, generation and emissions across the nation. The team built a dashboard to dynamically visualize the energy transition in 2010-24, and forecast 2025-30, by integrating EIA generation data with EPA emissions data across more than 30,000 generators nationwide.

Problem & Motivation

As the US energy grid is transitioning away from fossil fuels, there is no tool that offers narrative-driven insights and presents them in a coherent, user-driven interface. Crafting a narrative in a sea of energy data is challenging – yet it is critical for policymakers, investors, and project managers. This tool makes energy narrative creation and customization much more accessible so the relevant stakeholders can make better data driven decisions.



Data

Sources: EIA 860 data at the generator level and EPA eGRID data at the plant level (EIA 923 data was used as a sanity check). eGRID data was downcast to the generator level and joined with the EIA 860 data. The final data set contains 411K rows, 29 energy-related metrics, 16 other metrics over 15 years (107GB).

Collection and processing: The data was downloaded as Excel files. Data was cleaned, standardized, and joined in Python. The final dataset was stored as a CSV file to be used for aggregation, analytics calculation, clustering and forecasting.

Approach

Analytical Layer: Each state's energy transition metrics are dynamically calculated in D3.js. For each main metric and year, clustering of the 50 states is performed using that specific metric. Forecasts are created for the 2025-30 period using 5-year averages with 50% damping and 0 floors where relevant.

Visualization Layer: The focal point of the dashboard is the US state choropleth map. A metric dropdown allows the user to select one of 29 metrics to be displayed on the map. A time slider animates the map from 2010 to 2030 by progressing year-by-year, visualizing the selected map metric over time. The line chart below the map displays the time series of the metric at the national level. To the right of the map are three configurable state-level line charts that allow users to select state metrics for direct comparison across the entire metrics range.

Innovations: First, the project uses (messy) publicly available data to create a unique visual narrative with significant user interactivity. Second, an innovation is also the 2025-30 forecasting of energy metrics, something largely non-existent in other energy data dashboards. Third, K-means clusters are overlaid on the choropleth map, an innovation allowing the user to gain better insights compared to other existing approaches.

Experiment & Results

Evaluation: Five users performed 6 tasks using the team's dashboard as well as the EIA Electricity Data Browser and the EPA Data Explorer. Our solution enabled the fastest median response time of 36 seconds per question, 9 seconds better than the EPA and 11 seconds better than the EIA. From 0-100, the median accuracy of our users' responses was an impressive 100 vs EPA at 67 and EIA at 0. In terms of confidence of 0-100, our solution garnered 90 vs 70 for the EPA and 20 for the EIA. We also calculated system usability scales for the 3 dashboards based on answers from our users. Our solution scored 86 points (A-grade equivalent) compared to 54 for EPA ("D") and 33 for EIA ("F").

Results: We achieved our main goal: to create a narrative-driven visualization that combines historical analysis, forward-looking forecasts and an interactive user experience in a single interface. The successful conclusion of our project should add significant value to stakeholders: the ability to synthesize complex energy data into a coherent narrative and actionable insights; interactive visualizations; and a solid forecasting approach. Our work should help identify green-energy champion vs laggard states, uncover "hidden" trends and inform policy and capital allocation decisions at the local and state level.

Method Comparison: Compared to static charts or raw data tables, our D3.js choropleth map is highly interactive. The spatial-temporal approach allows users to easily view energy patterns that traditional methods miss. The 2025-30 forecasting and the embedded k-means overlay allows users to identify similar states across metrics and years much more efficiently.